DATA MANAGEMENT AND SHARING PLAN

Autonomous Multimodal Biological and Environmental Data Registry (AMBER)

1. Research Products and Data Organization

AMBER will produce three independently citable AI-ready datasets for forest bioresources, cropland food systems, and watershed water resources, connecting molecular, organismal, ecosystem, and landscape observations without assuming equivalent support or a universal latent space. A planned five foundational source families and up to ten additions will be reconciled against an approved source register; candidates include NEON, TRY, AmeriFlux, LTER/LTAR, NASA Earth observations, and path-specific genotype/phenotype resources. The register will distinguish providers, collections, products, assets, and observations, preserving identifiers, versions, coverage, rights, and citations. Authoritative observations will remain with providers.

Four product categories comprise integrated datasets; scientific metadata/provenance; numerical models/validation; and software/reproducible research. Records will distinguish source/observed, harmonized, derived, and AI-generated representations. Runs will carry incomplete, complete, or failed status; releases will distinguish generated fixtures, pilots, beta, and validated production products. Complete status establishes artifact completion, not scientific validity. Every release will identify custodians, schemas, source terms, checksums, and persistent identifiers.

2. Scientific Standards, Harmonization, and Validation

STAC Items/collections will describe spatiotemporal assets; CSV, Parquet/GeoParquet, JSON/JSONL, JSON-LD, and RDF/Turtle will carry metadata, graph exports, reviews, and lineage. COG/GeoTIFF, NetCDF/HDF5, and Zarr will remain provider-native access formats. Float32 NumPy .npy feature shards and non-pickled encoder arrays will accompany JSON manifests; other models may use Safetensors. Versioned JSON schemas, Python packages, OCI containers, and dependency lockfiles will define interfaces and environments.

Candidate semantic components include I-ADOPT for variable structure; SOSA/SSN, OBOE, and BCO for observations/sampling; ENVO/SWEET for environmental context; NCBITaxon, PCO, PO/UBERON, and OBA/PATO for entities/attributes; QUDT/UO for quantities/units; GeoSPARQL/OWL-Time for space/time; and RO/PROV-O for relations/derivation. Adoption remains evidence-reviewed. Mappings will retain source/target IDs, definitions, resource versions, evidence locations, direction, uncertainty, and reviewer/date. Equivalence, broader/narrower, part-of, spatial-temporal association, and model-proposed links will remain distinct. Unknown meaning will remain missing evidence.

Six scoped specialists will review semantics, units/calibration, support, rights, integrity, and quality against one frozen recipe digest. Findings use PASS, FAIL, NEEDS_EVIDENCE, or NOT_APPLICABLE. Deterministic consensus, QA synthesis of existing finding IDs, and exact-recipe human approval remain separate gates; disagreement or binding veto blocks execution. Scientific changes invalidate prior authorization. Models may select registered operations and schema-permitted parameters, never supply executable code, formulas, or authoritative constants. Tested deterministic software, including CubeDynamics through an external adapter where appropriate, executes approved recipes with inference off.

MOSAIKS will consume harmonized raster patches only; empirical filters and fitted normalization will use training observations exclusively. Other modalities require separately validated encoders. Verification will test checksums, dimensions, finite values, channel/unit/layout identity, duplicate observations, geographic/split leakage, and held-out performance against mean and shuffled-label baselines. Independent labels, proxies, and input-derived diagnostics remain distinct. Every vector will link to observation/window, applicable track/core recipe, operation/backend, encoder, agent findings, QA, human decision, source version, and rights. Metadata/window hashes will not imply whole-asset checksums.

3. Computational Infrastructure and Resource Management

ESIIL-OASIS will coordinate workflows/services; CyVerse VERDE-AI inference will use an isolated OpenClaw adapter with JSON-only model tasks, verified provider/model routes, and no model tools or silent fallbacks. Exact prompts/schemas, normalized responses, timestamps, available usage, failures, and vetoes will be preserved separately from operational logs. Secrets will remain outside public records. MESA discovery and iRODS policy integration will require verified interfaces and deployment agreements; private DuckDB metadata will remain locally controlled.

ACCESS CPU/GPU resources will support processing, fitting, and verification; Jetstream2 will support development and interactive research. A provisional 350,000-600,000 standard CPU SU allowance over three years covers a defined development/notebook envelope (1 SU/vCPU-hour), not all production, persistent services, or GPU work. Pilot timings will determine resource-specific Maximize requests and annual renewals, including cores/nodes, accelerator type, runtime, RAM, concurrency, source reads, and rebuilds. Shared OASIS contributions and cohort use will be reconciled to prevent double-counting.

Approved bounded windows will stream through harmonization directly into encoders without raw/harmonized disk caches, including tests. Offline tests use generated arrays/iterators. Restart records retain cursors, identities, manifests, and partial feature shards, not source/intermediate arrays. Storage will distinguish worker RAM, temporary feature staging, active workspace, distribution, and independent preservation.

4. Public Access, Sharing, and Release Schedule

NDP/OASIS catalogs and APIs will expose discovery and interpretive context; OSN S3-compatible object storage will distribute immutable, selectively retrievable feature partitions. Releases will include vectors, records.jsonl, harmonization_context.json, manifest.json, encoder filters/configuration/training context, verification labels/results, dataset cards, and reproducible examples. Cards will specify observation units, dimensions, precision, spatial/temporal coverage, uncertainty, limits, licenses, attribution, and permitted derivatives. Public accessibility will not imply redistribution permission; representations retain source-derived restrictions. FAIR and applicable CARE practices will govern sensitive locations, privacy, Indigenous authority, and rights review.

Year 1 delivers schemas, software, pillar pilots, and measured resource benchmarks; Year 2 delivers three beta datasets before synthesis use; Year 3 delivers revised validated releases. Cohort feedback will generate tracked issues, renewed review, affected-branch regeneration, and new versions without overwriting cited releases. Publication-supporting data/materials will be accessible at publication, with justified restrictions documented under NSF policy; persistent identifiers will enter NSF reporting.

5. Archiving, Preservation, and Retention

AMBER will request up to 50 TB standard OSN pilot capacity; larger production storage requires a separately approved OSN/pod or partner arrangement. Preliminary Year 3 scenarios allow 300-500 TB production, 20-40 TB shared workspace, and 300-500 TB additional preservation capacity (620-1,040 TB logical total), not confirmed allocations. Product sizing will use N x dimensions x bytes/feature x retained versions; lineage, labels, indexes, models, copies, duration, and physical protection overhead will be accounted separately. Jetstream2's general 20 TB Accelerate/40 TB Maximize storage limits require explicit workspace approval.

NDP will support discovery, OSN active distribution, and proposed TACC Corral/Ranch arrangements persistent collections/archive. OSN is not backed up; neither TACC access nor ten-year retention is automatic. AMBER will secure capacity, eligibility, funding, custodianship, and at least ten-year retention agreements before release, with an alternative approved archive if necessary. Deposits will retain versioned manifests, rights, recipes, review evidence, encoders, verification, and software environments; fixity and restoration will be tested. Persistent provider versions will be checked; unavailable source evidence requires authorized preservation or an approved exception.

6. Personnel, Accountability, and Governance

PI Amaral oversees compliance, governance, agreements, and preservation; Co-PI Ty Tuff oversees integrated datasets, metadata/provenance, software, and infrastructure; Co-PI Roth oversees encoders and validation. Three existing postdocs, one per dataset, lead source selection, scientific review, and release approval; two shared technical staff (data scientist/software developer) implement pipelines and services. Across 150 participants, three governance groups (Years 1-2) will curate semantics/rules and assess pilots; six synthesis groups (Years 2-3) will test real projects and return reproducible analyses and failure reports. The Year 2/Q2 summit will train over 100 participants. GitHub will version public decisions/code; restricted evidence and large arrays remain in controlled/object storage. Named stewards will reconcile allocations, consumption, permissions, release readiness, recovery, and retention at least annually.